

Voxel-based 3D Reconstruction from Multi-view Images

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1 Objective

The goal of this project is to reconstruct a volumetric (voxel-based) 3D representation of an object using multiple images captured from different viewpoints. Additionally, the project investigates how the number of training iterations affects the reconstruction quality and determines the optimal stopping point for training.

2 Theoretical Background

This work is based on the principles of volumetric rendering and 3D reconstruction from multi-view images. Unlike mesh-based approaches, volumetric models represent objects as continuous density fields in 3D space. Key concepts used in this project include:

- **Volumetric representation:** The object is represented as a 3D grid of voxels, each storing density and color.
- **Ray sampling:** Rays are cast from the camera through the volume, sampling points along each ray.
- **Volume rendering:** The final image is obtained by integrating density and color along each ray.
- **Silhouette supervision:** Binary masks help the model learn the object shape.
- **Multi-view consistency:** The model learns a consistent 3D structure from multiple 2D views.

3 Dataset Preparation

Instead of using the default synthetic cow from the textbook, a new object (**a rubber duck model**) was used.

Steps performed:

1. A 3D .obj model with texture was loaded.
2. The model was normalized and reoriented to match the coordinate system.
3. A multi-view dataset was generated by rotating the camera around the object.
4. For each view:
 - RGB image was rendered
 - Silhouette mask was extracted

The dataset consisted of:

- 40 RGB images
- 40 silhouette images

Duck Preview



Figure 1: Duck Preview



Figure 2: Duck RGB Dataset

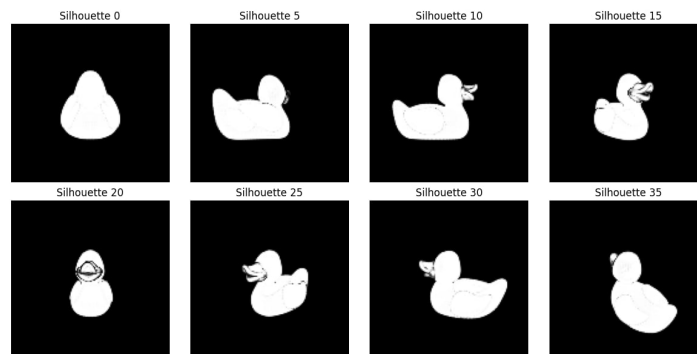


Figure 3: Duck silhouette Dataset

4 Model Architecture

A voxel-based neural representation was implemented using a learnable 3D grid.

The model parameters include:

- **Density grid:** determines the occupancy of each voxel
- **Color grid:** stores RGB values

The model outputs:

- Density values in the range $[0, 1]$
- Color values in the range $[0, 1]$

A differentiable volume renderer was used to produce images from arbitrary viewpoints.

5 Training Process

The model was trained using the following loss function:

- **Silhouette loss (MSE)** - enforces correct object shape
- **RGB reconstruction loss (MSE)** - enforces color consistency
- **Density regularization** - suppresses noise in empty space

Final loss function:

$$\text{Loss} = 2.0 \cdot L_{\text{sil}} + 1.0 \cdot L_{\text{rgb}} + 0.01 \cdot L_{\text{reg}}$$

where:

- L_{sil} — silhouette loss
- L_{rgb} — RGB reconstruction loss
- L_{reg} — mean density penalty

Training was performed with:

- Grid size: 32^3
- Learning rate: 0.05
- Ray samples per ray: 64

6 Experimental Setup

Three experiments were conducted with different numbers of training iterations:

- 150 iterations
- 300 iterations
- 600 iterations

During training, intermediate snapshots were saved every 50 iterations to visualize the learning process. These snapshots were later combined into GIF animations.

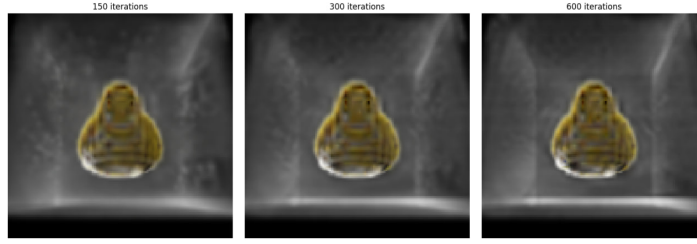


Figure 4: Duck RGB Experiments

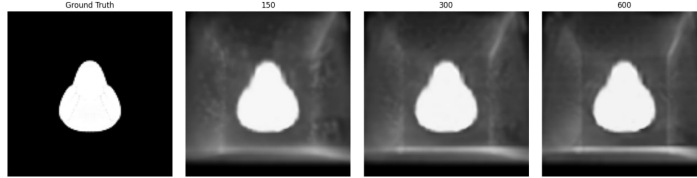


Figure 5: Duck Silhouette Experiments

7 Results

7.1 Loss Analysis

The training loss decreased rapidly during the first 100 iterations, indicating fast learning of the object structure. After approximately 150–200 iterations, the loss curve began to flatten.

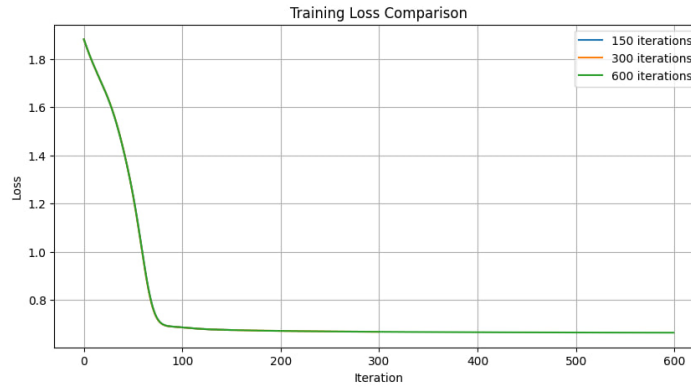


Figure 6: Duck Loss

Observations:

- 150 iterations: good initial reconstruction, but still unstable
- 300 iterations: stable and accurate reconstruction
- 600 iterations: minimal improvement compared to 300

This shows that most of the learning occurs early in training.

7.2 Visual Results

The reconstructed object was successfully recovered from different viewpoints. The model captured:

- overall shape of the duck

- basic color distribution
- approximate geometry

However, some artifacts were observed:

- slight blurring due to volumetric rendering
- low-density noise in empty space (visible as a “transparent cube”)
- limited detail due to voxel resolution

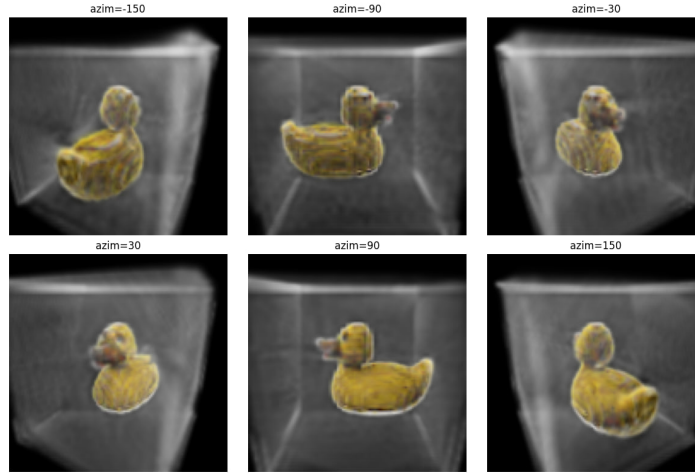


Figure 7: Final Duck visualization

7.3 Effect of Regularization

Adding density regularization improved the results by:

- reducing noise in empty space
- making the object more distinct

However, too strong regularization caused:

- loss of object structure
- overly transparent reconstructions

This demonstrates the importance of balancing loss components.

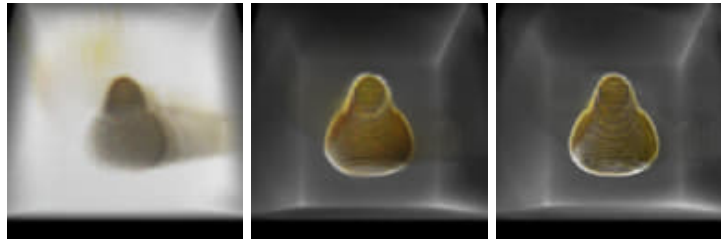


Figure 8: Duck (Strong Reg, high density, Ray)

8 Conclusion

In this project, a voxel-based 3D reconstruction system was successfully implemented and trained using multi-view images.

Key conclusions:

- The model can reconstruct a recognizable 3D object from 2D images
- Most learning occurs within the first 150–300 iterations
- Further training yields diminishing returns
- The optimal number of iterations is approximately 300
- Proper regularization is critical for removing noise

This work demonstrates the effectiveness of volumetric rendering techniques for 3D reconstruction tasks.